

COLLEGE OF ENGINEERING – DEPARTMENT OF COMPUTER SCIENCE

CS 4176	Senior Project 1	Assignment 01	
CLO No.	CLO 1	Total Marks	5
	Literature Review		
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Task Statement:

Students are required to conduct a structured and critical literature review relevant to their proposed Senior Project. The review must survey existing research, technologies, and methodologies closely aligned with the project's problem domain and objectives.

The literature review must include research papers published in reputed and peer-reviewed venues (e.g., well-known journals, conferences, or publishers). The majority of the selected papers should be recent, reflecting the current state of the art in the field. Each paper must be clearly relevant to the project and demonstrate academic credibility.

Students are expected to:

- Identify and justify the selection of reputable and recent sources.
- Summarize the core ideas, methods, and findings of each work.
- Critically analyze similarities, differences, strengths, and limitations across the reviewed studies.
- Clearly position their proposed project within the existing literature, highlighting gaps, limitations, or open problems that the project aims to address.

The literature review should be written in a formal academic style, properly cited, and structured in a logical manner that supports the motivation and direction of the proposed project.

Submission:

- Only one group member should submit this document.
- Names of all group members must be clearly mentioned at the top of the document.

EpiLoad: An Integrated AI Framework for Hospital Load Management

1. Introduction

Epidemic and pandemic outbreaks place significant pressure on healthcare systems, often exposing limitations in hospital capacity, patient flow, and inter-facility coordination [1], [3], [5]. Sudden increases in patient volume during infectious disease outbreaks can overwhelm emergency departments, saturate intensive care units, and disrupt routine healthcare delivery [1], [10]. These challenges were clearly observed during the COVID-19 pandemic, where hospitals across different healthcare systems experienced shortages of beds, staff, and critical resources despite varying levels of preparedness [3], [5], [20].

Hospital load management refers to the ability of healthcare institutions to monitor, allocate, and utilize resources such as beds, staff, and equipment while maintaining safe and timely patient care [10], [11]. Under outbreak conditions, this process becomes more complex due to uncertainty in demand, rapid changes in patient acuity, and constraints on workforce availability [3], [13]. When hospital load is not effectively managed, it can result in prolonged waiting times, delayed admissions, increased mortality risk, and reduced quality of care [10], [27]. For this reason, improving load management and coordination has become a key operational concern in outbreak response planning [20], [21].

Advances in data analytics and artificial intelligence have created new opportunities to support hospital operations under high-demand conditions [13], [19]. Predictive models can be used to estimate future admissions and resource utilization, while optimization and simulation approaches provide structured methods for planning and decision-making [2], [17]. In addition, digital command centers and regional coordination frameworks have been introduced to improve visibility and facilitate patient transfers across healthcare networks [11], [12]. While these developments are promising, their practical impact depends on how effectively they can be integrated into real operational workflows.

Despite the range of approaches proposed in the literature, there is considerable variation in scope, methodology, and applicability [13], [12]. Some studies focus on hospital-level decision-making, while others examine regional or system-wide coordination [15], [21]. At the same time, challenges related to data quality, interoperability, ethical constraints, and real-time implementation continue to limit the adoption of

advanced analytical tools in practice [8], [22]. This suggests that existing solutions may not fully address the complexity of real-world hospital operations during epidemic surges.

The objective of this literature review is to examine existing research on hospital load management and coordination during epidemic outbreaks, with particular emphasis on data analytics and artificial intelligence approaches. The review considers operational challenges, capacity management strategies, predictive and prescriptive analytical methods, and cross-hospital coordination mechanisms reported in the literature. By synthesizing findings across these areas, the review aims to identify key limitations in current approaches and position the proposed project within the broader research landscape.

The remainder of this paper is organized as follows. Section 2 describes the methodology used to select and screen relevant literature. Section 3 introduces key background concepts and definitions. Sections 4 and 5 present the thematic review and comparative analysis of existing studies. Section 6 discusses implementation challenges and open issues, Section 7 positions the proposed project, and Section 8 concludes the paper.

2. Methodology for Literature Selection

This literature review adopts a structured approach to identify relevant and credible studies related to hospital load management and coordination during epidemic outbreaks. Academic sources were collected from established scholarly databases, including IEEE Xplore, ScienceDirect, and Google Scholar. These platforms were selected because they provide broad access to peer-reviewed research in computer science, healthcare systems, operations research, and medical informatics.

The search process used combinations of keywords related to epidemic outbreaks, hospital capacity management, patient flow, resource allocation, command centers, and artificial intelligence techniques. Variations and synonyms of these terms were also included to capture a wider range of relevant studies. The review focuses primarily on literature published between 2020 and 2025, reflecting recent developments and insights from large-scale outbreak events, particularly the COVID-19 pandemic.

Only peer-reviewed journal articles and conference papers were included as core sources. A limited number of authoritative reports and preprints were also considered when they provided useful policy perspectives or emerging methodological contributions. Studies that focused exclusively on clinical diagnosis, epidemiological transmission modeling, or laboratory-based research without direct relevance to hospital operations or capacity management were excluded.

Following the initial search, the identified studies were screened based on relevance, publication quality, and alignment with the objectives of this review. The final selection prioritizes work that addresses operational stress, hospital capacity and patient flow management, predictive analytics, optimization techniques, and cross-hospital coordination. This selection process helps ensure that the reviewed studies are directly aligned with the scope and objectives of the proposed project.

3.1 Epidemic and Outbreak Surges in Healthcare Systems

Epidemic and outbreak surges describe periods of rapid and sustained increases in healthcare demand driven by infectious disease events [1], [3]. Unlike routine seasonal variations, these surges involve sudden spikes in patient arrivals, longer hospital stays, and higher levels of clinical severity [5], [16]. Such conditions can quickly push healthcare systems beyond their normal operating capacity, creating significant operational strain [1], [20].

During large-scale outbreaks such as COVID-19, healthcare facilities faced simultaneous pressure across emergency departments, inpatient wards, and intensive care units [1], [5]. The literature indicates that these surge conditions disrupt established workflows, reduce system flexibility, and reveal limitations in hospital preparedness [3], [13]. In addition to increased patient volume, outbreaks also affect staffing availability, equipment usage, and coordination between departments [3], [10]. This broader impact highlights that epidemic surges function as system-wide stress tests rather than isolated clinical events [20], [21].

3.2 Hospital Load, Capacity, and Patient Flow

Hospital load extends beyond simple bed occupancy and reflects the overall pressure placed on healthcare systems. It includes demand on physical resources such as beds, ventilators, and treatment spaces, as well as human resources including physicians, nurses, and support staff, alongside the efficiency of operational processes [10], [11]. During outbreak conditions, this pressure intensifies due to longer lengths of stay, delayed discharges, and increased patient acuity [16], [25].

Closely related to this is hospital capacity, which represents the ability of a healthcare facility to accommodate and treat patients within its available resources [3], [27]. Capacity can be understood in terms of structural capacity (beds and infrastructure), workforce capacity (staff availability and skill mix), and functional capacity, which reflects how effectively these resources are utilized [27], [11]. In many surge scenarios, functional capacity becomes the limiting factor, as coordination inefficiencies or delays in patient movement reduce the effective use of available resources [11], [26].

Another key concept is patient flow, which describes how patients move through the healthcare system from arrival and triage to admission, treatment, transfer, and discharge [27], [26]. Maintaining efficient patient flow is essential during high-demand periods, as disruptions can quickly lead to bottlenecks across the system. Poor flow management is commonly associated with emergency department crowding, delays in inpatient placement, and restricted access to critical care services [25], [24]. For this reason, patient flow management remains a central focus in operational and health informatics research on outbreak preparedness [11], [13].

3.3 Analytics and Decision-Support Approaches in Healthcare Operations

To address surge-related challenges, the literature increasingly emphasizes the role of data analytics and decision-support systems in healthcare operations [13], [19]. These approaches are commonly categorized into descriptive, predictive, and prescriptive analytics, each contributing differently to operational decision-making [13], [12].

Descriptive analytics provides a foundation for situational awareness by summarizing and visualizing historical and real-time data [11], [12]. In hospital environments, this typically appears in the form of dashboards, occupancy reports, and command center displays that track patient volumes, bed availability,

and resource utilization. While these tools improve visibility and support monitoring, they do not directly inform future actions.

Building on this, predictive analytics focuses on estimating future system states using historical data and current trends [2], [16]. In outbreak contexts, predictive models are applied to forecast patient arrivals, intensive care demand, length of stay, and overall resource requirements [2], [17]. These forecasts allow hospitals to anticipate potential stress points and prepare responses in advance. However, their effectiveness depends heavily on data quality and their ability to adapt to rapidly changing outbreak conditions [22].

Extending beyond prediction, prescriptive analytics introduces decision-support mechanisms that recommend specific actions to improve outcomes [14], [17]. This includes optimization models, simulation-based planning, and systems that evaluate alternative strategies for resource allocation, patient transfers, and surge response planning [14], [15]. Such approaches are particularly important in multi-hospital settings, where coordinated decision-making is required. Despite their potential, these systems face practical challenges related to interpretability, integration into clinical workflows, and user trust [8], [22].

4. Literature Review (Thematic Findings)

4.1 Operational Stress in Hospitals During Epidemic Outbreaks

Epidemic outbreaks place healthcare systems under sudden and sustained operational pressure that extends far beyond routine fluctuations in patient demand. Carenzo et al. [1] document how COVID-19 surges led to rapid increases in hospital admissions and prolonged lengths of stay, particularly in critical care settings, while Gupta et al. [3] emphasize that these pressures are further intensified by workforce limitations and resource constraints. Unlike seasonal variations, these surges occur within compressed timeframes, creating simultaneous strain across multiple hospital units. This suggests that epidemic-related stress is driven not only by volume, but also by the speed and complexity with which demand escalates.

One of the most immediate manifestations of this pressure appears in emergency department congestion. Chang et al. [24] show that increased patient arrivals during outbreaks lead to delays in triage and admission processes, while Farimani et al. [25] highlight how these delays propagate through the system, contributing

to overcrowding and extended waiting times. As patient flow slows, emergency departments become bottlenecks that affect both epidemic and non-epidemic cases. Franklin et al. [27] further indicate that delays in transferring patients from emergency departments to inpatient units significantly disrupt overall care coordination. This is important because it shows that congestion is not isolated to a single unit, but instead creates system-wide inefficiencies.

Pressure becomes even more pronounced in intensive care units during epidemic events. Carenzo et al. [1] report that COVID-19 outbreaks led to rapid ICU saturation, particularly for patients requiring advanced respiratory support, while Pett et al. [5] describe how critical care transfer networks were required to redistribute patients when local capacity was exceeded. Gupta et al. [3] add that ICU capacity is constrained not only by physical beds, but also by staffing availability and specialized expertise. While Carenzo et al. [1] focus on the physical and clinical demands placed on ICUs, Gupta et al. [3] emphasize workforce-related limitations, highlighting that capacity is both a structural and organizational challenge. This means that expanding ICU capacity during crises is significantly more complex than simply increasing bed numbers.

Beyond emergency and critical care units, hospital-wide bed shortages further intensify operational stress. Franklin et al. [27] show that high occupancy levels reduce flexibility in patient placement, while Hadinejad et al. [10] explain that disruptions in patient flow can lead to delays across departments. Al-Harbi et al. [26] further demonstrate that discharge delays often result from limited availability of downstream services such as rehabilitation or community care. As a result, clinically stable patients remain in acute-care beds longer than necessary, creating bed-blocking effects that cascade throughout the hospital system.

Another key dimension of operational stress is staffing pressure. Gupta et al. [3] highlight how epidemics increase staff absenteeism due to illness, quarantine, and burnout, reducing available workforce capacity at the same time that patient demand rises. Abir et al. [20] show that hospitals often respond by implementing emergency staffing models, extending shifts, and redeploying personnel across departments. While these strategies help maintain operations in the short term, they also increase fatigue and reduce long-term resilience. This reinforces the idea that workforce capacity is a central limiting factor in outbreak response.

Evidence from large-scale outbreaks consistently shows that operational stress is not solely a function of patient volume, but rather the interaction between demand, resource availability, and coordination mechanisms. Carenzo et al. [1] and Pett et al. [5] illustrate how hospitals struggled when demand exceeded local capacity, while Mebrahtu et al. [11] show that a lack of real-time visibility into capacity and patient

flow made coordination more difficult. In contrast, Mebrahtu et al. [11] and Feda et al. [12] demonstrate that hospitals with structured monitoring systems and centralized coordination were better able to manage surge conditions. This comparison suggests that coordination and visibility play a critical role in determining how effectively hospitals respond to operational stress.

Overall, the literature indicates that operational stress during epidemic outbreaks emerges as a multidimensional challenge involving emergency congestion, critical care saturation, bed shortages, and workforce constraints. What stands out is that these factors do not operate independently, but instead reinforce each other under surge conditions. This is important because it highlights the need for integrated approaches that move beyond reactive crisis management toward proactive capacity awareness and coordinated response mechanisms.

4.2 Hospital Capacity and Patient Flow Management

Research on hospital capacity management increasingly emphasizes that capacity is not only a matter of bed count, but also of how quickly patients can move through the system. Hadinejad et al. [10] show in their scoping review that patient flow during biological events depends on coordinated bed management, triage efficiency, discharge processes, and inter-unit communication, which means that hospitals can appear to have structural capacity while still operating under severe functional bottlenecks. This broader framing is important because it shifts attention from static occupancy to operational throughput, which is more relevant during epidemic surges.

Several studies examine how hospitals improve flow through centralized coordination structures. Franklin et al. [27] report that hospital capacity command centers are designed to provide centralized visibility over occupancy, transfers, and throughput constraints, while Mebrahtu et al. [11] evaluate command-centre implementation in the UK NHS and link it to improved data quality and patient-flow oversight. Feda et al. [12], in a 2025 scoping review of digital hospital command centers, similarly argue that these systems can reduce crowding and improve emergency department throughput when supported by interoperable data and clear governance. Taken together, Franklin et al. [27], Mebrahtu et al. [11], and Feda et al. [12] suggest that command centers are strongest when they function as active operational coordination tools rather than

passive dashboards. However, these studies also show that command centers do not solve bottlenecks automatically; their impact depends heavily on timely data, staff adoption, and institutional alignment.

A more direct operational contribution comes from Bishop et al. [9], who move beyond descriptive monitoring and apply machine learning to predict which inpatients are likely to be ready for discharge within the next 24 hours using electronic health record data from Oxford University Hospitals. Their proposed workflow ranks all inpatients by discharge likelihood and flags the top 20% for clinician review. Performance was strongest for patients early in admission, with positive predictive value reaching 0.96 for planned patients and 0.94 for emergency patients on day 1, but declining to 0.66 and 0.71, respectively, for patients still admitted after 7 days. This study is highly relevant because it connects prediction to a concrete patient-flow intervention: faster identification of discharge candidates to relieve bed pressure during crisis periods. Its main limitation is that it supports discharge prioritization only, not full hospital-wide load balancing or inter-hospital coordination.

Al-Harbi et al. [26] complement this line of work by showing that structured case-management practices can improve operational efficiency and reduce avoidable flow delays. Their contribution is less algorithmic than Bishop et al. [9], but it is important because it highlights that even strong predictive models will have limited impact if discharge workflows, case review, and downstream placement processes remain inefficient. While Bishop et al. [9] focus on patient-level discharge prediction to improve bed turnover, Franklin et al. [27], Mebrahtu et al. [11], and Feda et al. [12] emphasize system-level coordination through command centers and dashboards. This contrast shows that effective hospital flow management requires both micro-level prediction and macro-level operational oversight. In other words, predictive discharge tools and process redesign should be viewed as complementary rather than competing approaches.

Li et al. [6] extend the capacity-management discussion into emergency overflow environments by examining artificial intelligence applications in Fangcang shelter hospitals, which are rapidly deployable, modular facilities used to isolate and manage large patient populations during outbreaks. Rather than focusing on a single prediction model, they identify several operational AI use cases, including site selection and design optimization, clinical decision support, patient engagement and documentation, robotics, and operational management. This paper is valuable for the review because it broadens the discussion of capacity management beyond conventional hospitals and shows how surge infrastructure itself can be designed as an intelligent operational environment. At the same time, Li et al. [6] also highlight serious barriers, including limited data access, privacy and governance constraints, weak algorithmic adaptability in

fast-changing emergency settings, system fragmentation, high costs for temporary facilities, and hardware reliability issues. That makes the paper useful not because it proves deployment success, but because it shows what is required for AI-enabled overflow capacity to be realistic.

Overall, Franklin et al. [27], Mebrahtu et al. [11], and Feda et al. [12] show that visibility tools improve coordination, while Bishop et al. [9] and Al-Harbi et al. [26] show that discharge support and workflow redesign improve throughput more directly. Li et al. [6] extend this discussion by showing that surge capacity may also depend on intelligent temporary-care infrastructure during large outbreaks. However, these components are still mostly studied separately rather than as parts of a unified hospital and network-level load-management framework. That gap directly motivates the proposed project.

4.3 Predictive Analytics for Hospital Demand and Occupancy

Predictive analytics is one of the most developed research streams in hospital load management, but the studies differ substantially in what they predict and how useful their outputs are in practice. Wohlfender et al. [16], for example, focus on short-term forecasting of COVID-19 hospital admissions using routine patient data from six hospitals in the Bern region of Switzerland between February 2020 and June 2023. Their models generate forecasts one to five weeks ahead, which makes them suitable for short-term operational planning. In contrast, Garber and Okhrin [2] focus on ICU length-of-stay prediction within a simulation-based bed-capacity framework, while Mahmoudian et al. [18] estimate future bed capacity requirements using machine learning and deep learning approaches. Bishop et al. [9], on the other hand, deal with discharge readiness at the individual patient level. This suggests that predictive analytics is not addressing a single problem, but several different operational needs within hospital systems.

Looking more closely at Wohlfender et al. [16], their main strength lies in anticipating demand before hospitals become visibly overloaded. This is particularly useful in practice, since hospital administrators often need early signals to prepare for incoming surges. A limitation is that the study stops at prediction and does not explain how hospitals should respond once increased demand is identified. In that sense, its value is mainly in early warning rather than direct decision support.

Garber and Okhrin [2] approach the problem differently by focusing on ICU length of stay, which directly affects how long beds remain occupied. They use the HiRID dataset, which contains high-frequency ICU data for more than 33,000 patients, and compare models such as logistic regression and XGBoost. This makes their work more relevant for internal capacity management. It is less focused on admission forecasting alone. Compared with Wohlfender et al. [16], who focus on incoming demand, Garber and Okhrin [2] deal with resource usage after admission. One drawback is that their approach is still limited to single-hospital settings and does not extend to coordination across multiple facilities.

Mahmoudian et al. [18] go further by linking prediction to capacity planning decisions. Their study combines LOS classification methods, including support vector machines and decision trees, with forecasting approaches such as SARIMA and LSTM. Using 51,231 patient records from a heart ward, they estimate that bed capacity would need to increase from 45 to 137 beds by 2026. This is a strong example of translating predictive outputs into planning decisions. That said, the study remains focused on a single ward and does not address real-time decision-making or system-wide coordination, which limits its applicability in large-scale surge scenarios.

Bishop et al. [9] focus on a more immediate operational issue by predicting discharge readiness within the next 24 hours. Their model ranks patients and identifies those most likely to be discharged, with positive predictive values of 0.96 for planned admissions and 0.94 for emergency admissions on the first day, although performance declines for longer stays. Compared with Mahmoudian et al. [18], which operate at a planning level, Bishop et al. [9] provide outputs that can be used directly in daily hospital operations. A limitation is that their approach still depends on clinician validation and does not address broader allocation or coordination challenges.

When these studies are considered together, it becomes clear that predictive analytics operates at multiple levels within hospital systems. Wohlfender et al. [16] support early surge detection, Garber and Okhrin [2] contribute to ICU capacity management, Bishop et al. [9] improve discharge processes, and Mahmoudian et al. [18] support longer-term planning. This is important because each study solves a different part of the problem, which makes it difficult to rely on any single approach in practice. A model may perform well technically but still have limited impact if it is not connected to concrete operational decisions.

A consistent limitation across these studies is that they stop at prediction. Mahmoudian et al. [18] estimate capacity needs but do not address how hospitals should respond when those limits are reached. Bishop et al.

[9] identify discharge opportunities but do not optimize transfers or system-wide allocation. Similarly, Wohlfender et al. [16] and Garber and Okhrin [2] provide useful forecasts, yet neither offers a complete decision-support framework. This also means that prediction alone is not sufficient, especially in the context of epidemic surges where coordination between hospitals becomes critical.

Overall, the literature shows that different predictive targets serve different operational purposes. Admissions forecasting helps anticipate surges, LOS and discharge prediction support bed turnover, and capacity forecasting supports planning. However, these elements are still mostly studied separately. There is still no integrated system that combines prediction with real-time decision-making and cross-hospital coordination, which is the gap this project aims to address.

4.4 Optimization and Simulation for Surge Response

While predictive analytics estimates future hospital demand, optimization and simulation models aim to determine how resources should be allocated once that demand is anticipated. Abdel-Rahman et al. [7] and Leefink et al. [14], for example, focus on transforming predicted demand into allocation strategies that guide how beds, staff, and transfers should be managed during epidemic surges. These approaches move beyond forecasting and into prescriptive decision support, providing structured mechanisms for managing limited resources under high-pressure conditions.

Optimization models are designed to identify efficient allocation strategies under defined constraints. In outbreak scenarios, these constraints typically include ICU bed availability, ventilators, trained personnel, and inter-hospital transfer capacity [7]. Leefink et al. [14] apply mathematical programming techniques to determine how patients should be distributed across facilities in order to balance occupancy levels, while Parker et al. [22] focus on minimizing overload and reducing transfer delays at a regional level. This makes optimization particularly relevant when demand exceeds local capacity and coordinated redistribution becomes necessary. At the same time, Pett et al. [5] highlight that real-world transfer decisions often depend on operational and logistical factors that are not always captured in formal models.

Another important direction involves dynamic bed allocation strategies. Rather than treating capacity as fixed, Leeftink et al. [14] and Parker et al. [22] show that allocation decisions can be updated over time as patient inflow and discharge patterns evolve. This is especially useful during multi-wave epidemics, where demand fluctuates and resource requirements change rapidly. By incorporating time-dependent variables, Abdel-Rahman et al. [7] provide a more realistic representation of hospital operations under changing conditions. This suggests that static allocation models may be insufficient in environments where demand is highly volatile.

Simulation techniques complement optimization by allowing hospitals to evaluate potential strategies before implementation. Farimani et al. [25] use discrete-event simulation to replicate patient flow processes and assess how operational changes affect system performance, while Feda et al. [12] emphasize the role of simulation in testing policy decisions within digital hospital command centers. Through scenario-based analysis, these models allow administrators to explore the impact of increasing ICU capacity, accelerating discharge procedures, or modifying triage protocols. This is particularly valuable in practice because real-world experimentation during outbreaks is often infeasible.

A further development involves integrating optimization with predictive forecasting. Abdel-Rahman et al. [7] and Parker et al. [22] demonstrate how forecasted demand can be used as input for allocation algorithms, creating more adaptive decision-support systems. This allows hospitals not only to anticipate surges but also to generate actionable plans in response to projected demand levels. However, this integration also introduces additional complexity, as it requires reliable data pipelines and coordination across departments and institutions.

Despite their potential, these approaches face several limitations. Abdel-Rahman et al. [7] and Feda et al. [12] note that many models rely on assumptions such as complete data availability or instantaneous transfer capability, which may not reflect real-world constraints. Leeftink et al. [14] further highlight that optimization models often prioritize efficiency without fully accounting for ethical considerations or equity in resource allocation. In contrast, Pett et al. [5] show that real-world decisions during epidemic surges must balance operational efficiency with clinical judgment and regulatory requirements. This indicates that purely technical models may not fully capture the complexity of decision-making in crisis settings.

Overall, optimization and simulation research provides structured tools for surge response planning by translating predictive insights into allocation strategies. Abdel-Rahman et al. [7] and Leeftink et al. [14]

demonstrate the potential of these methods to improve resource utilization, while Parker et al. [22] highlight their value at a regional coordination level. However, their effectiveness depends on realistic assumptions, interoperability across systems, and alignment with clinical and governance frameworks. This reinforces the need for approaches that combine predictive, operational, and coordination capabilities within a unified framework.

4.5 Cross-Hospital Coordination and Regional Load Management

Epidemic surges rarely affect hospitals in isolation. Demand often fluctuates unevenly across regions, creating situations in which some facilities experience critical overload while others retain unused capacity [5], [12]. This imbalance has led to increased research focus on cross-hospital coordination and regional load management as mechanisms for distributing patient demand more effectively. Mitchell et al. [4], for example, examine how regional coordination strategies can support load balancing across hospital networks during periods of extreme demand. This suggests that managing surge response at a system level is necessary when local capacity becomes insufficient.

At the regional level, coordination frameworks are designed to enable patient transfers, shared capacity visibility, and centralized decision-making across multiple hospitals. Rather than treating hospitals as independent units, these models approach the healthcare system as an interconnected network [5]. Mitchell et al. [4] show that structured transfer protocols can reduce imbalance by redistributing patients from overloaded facilities to those with available capacity, while Feda et al. [12] highlight the importance of centralized visibility through digital coordination platforms. This is important because it shifts the focus from local optimization to system-wide stability.

Practical implementations of this approach can be seen in transfer centers and regional command hubs. Franklin et al. [27] report that these systems aggregate real-time data on bed availability, ICU occupancy, and transfer requests across multiple institutions, while Feda et al. [12] emphasize that standardized communication processes reduce delays caused by fragmented coordination. During large-scale outbreaks, Pett et al. [5] show that centralized coordination was associated with more effective redistribution of

critically ill patients. Compared with ad hoc hospital-to-hospital negotiations, these structured approaches provide more consistent and timely decision-making.

Optimization-based regional models extend this concept by introducing formal allocation strategies. Abdel-Rahman et al. [7] and Parker et al. [22] develop models that determine how patients should be distributed across geographic networks based on factors such as transportation time, resource availability, and projected demand. While these models aim to minimize overload or reduce mortality risk, their implementation depends heavily on real-world constraints. In contrast to these theoretical models, Pett et al. [5] show that operational decisions during outbreaks often involve logistical and clinical considerations that are not fully captured in optimization frameworks. This indicates that mathematical models alone may not be sufficient without integration into practical coordination systems.

Despite the demonstrated potential of regional coordination, several challenges remain. Gupta et al. [3] highlight that hospitals often operate under different administrative structures and data systems, which limits seamless information exchange. Feda et al. [12] further note that coordination mechanisms depend on timely and accurate data, which may become unreliable during crisis conditions. Ethical considerations also arise when prioritizing transfers between regions, particularly when multiple areas face simultaneous resource shortages [5]. This means that coordination is not only a technical problem, but also an organizational and ethical one.

Overall, the literature shows that effective outbreak response requires coordination beyond individual hospital boundaries. Mitchell et al. [4] and Pett et al. [5] demonstrate that regional approaches improve load balancing and reduce localized overload, while Abdel-Rahman et al. [7] and Parker et al. [22] highlight the role of structured allocation strategies. However, these approaches still depend on interoperability, governance, and real-time data integration. This reinforces the need for systems that combine predictive, operational, and coordination capabilities within a unified framework.

5. Comparative Analysis of Existing Approaches

The reviewed literature presents three main analytical approaches for managing hospital load during epidemic outbreaks: predictive forecasting, optimization and simulation modeling, and cross-hospital coordination frameworks [2], [7], [5]. Although each approach addresses operational stress, they differ in terms of objectives, assumptions, and practical applicability. Understanding these differences is important for identifying how each contributes to surge management and where their limitations lie.

Predictive forecasting models are primarily concerned with estimating future demand. Garber and Okhrin [2] and Wohlfender et al. [16], for instance, focus on anticipating admissions, ICU occupancy, or length of stay within a defined time horizon. These models aim to answer how many patients are likely to require care, which makes them particularly useful for short-term planning. Statistical time-series approaches emphasize interpretability and trend analysis, while machine learning methods prioritize predictive accuracy and pattern detection. In practice, forecasting tools are valuable because they provide early signals of rising demand. However, they do not determine how resources should be allocated once capacity limits are reached. This means that their strength lies in anticipation rather than direct decision-making.

In contrast, optimization and simulation approaches move beyond prediction by providing structured guidance for resource allocation. Abdel-Rahman et al. [7] and Parker et al. [22] develop models that determine how beds, staff, and patient transfers should be distributed under constrained conditions, while Farimani et al. [25] use simulation techniques to evaluate the impact of different operational strategies. Compared with forecasting models, these approaches offer more direct decision support by linking system inputs to actionable outputs. However, their effectiveness depends on the validity of underlying assumptions, such as data completeness and transfer feasibility. In rapidly changing outbreak conditions, these assumptions may not always hold, which can limit the reliability of optimization-based decisions.

Cross-hospital coordination frameworks introduce a broader, system-level perspective. Rather than focusing on a single institution, these approaches address how demand can be redistributed across multiple hospitals. Mitchell et al. [4] and Pett et al. [5] show that coordinated transfer systems and regional collaboration can reduce localized overload and improve overall system resilience, while Gupta et al. [3] highlight the importance of governance and institutional alignment. Unlike optimization models, which focus on determining ideal allocation patterns, coordination frameworks deal with how those decisions are implemented in practice. This includes communication, data sharing, and regulatory considerations. As a

result, their effectiveness depends less on algorithmic design and more on operational feasibility and collaboration between institutions.

A key distinction across these approaches lies in their operational scale. Forecasting models are typically applied at the hospital or unit level, such as predicting ICU demand within a single facility [2]. Optimization models can operate at either the hospital or regional level depending on their design [7], while coordination frameworks inherently function across multiple institutions [5]. This difference in scale directly affects the complexity of implementation, as larger-scale systems require more extensive data integration and coordination mechanisms.

Another important difference relates to implementation maturity. Forecasting tools are increasingly integrated into hospital dashboards and analytics platforms [27], indicating that they are already part of routine decision-making processes. In contrast, optimization and simulation models are more often explored in research settings than fully deployed in practice [25], [7]. Coordination frameworks have shown practical value during large-scale outbreaks [5], but their success often depends on external factors such as policy alignment and administrative cooperation [3]. This suggests that practical feasibility, rather than theoretical strength, plays a major role in determining which approaches are adopted in real-world settings.

Across all three approaches, a recurring limitation is fragmentation. Many studies focus on a single analytical dimension without integrating forecasting, optimization, and coordination into a unified system [12], [22]. As a result, predictive insights may not translate into actionable allocation strategies, and allocation decisions may not align with regional coordination mechanisms. This disconnect reduces the overall effectiveness of individual approaches when applied in isolation.

Taken together, the literature shows that no single method is sufficient on its own. Forecasting supports early anticipation, optimization provides structured allocation strategies, and coordination enables system-wide resilience. The main challenge lies in combining these capabilities into integrated, interoperable systems that can adapt to rapidly evolving outbreak conditions. The key distinctions across these approaches are summarized in Table 1.

Table 1

Comparative Analysis of Surge Management Approaches in the Reviewed Literature

Approach Category	Primary Objective	Typical Methods	Operational Scale	Strengths	Key Limitations	Implementation Status
Predictive Forecasting	Estimate future hospital demand and occupancy	Time-series models, regression models, machine learning algorithms (e.g., [2], [16])	Primarily hospital-level	Improves short-term anticipation of admissions, ICU demand, and discharge patterns; supports proactive planning	Does not prescribe allocation decisions; effectiveness depends on data quality and rapidly changing outbreak conditions	Increasingly integrated into hospital dashboards and analytics systems [27]
Optimization and Simulation	Allocate limited resources efficiently under constraints	Linear and integer programming, dynamic optimization, discrete-event simulation (e.g., [7], [22], [25])	Hospital-level and regional-level	Provides structured allocation strategies and enables scenario testing under different demand conditions	Often relies on simplifying assumptions (e.g., transfer feasibility, data completeness); limited real-world deployment	Predominantly research-oriented, with partial operational adoption

Cross-Hospital Coordination	Balance load across multiple institutions	Transfer protocols, regional command centers, shared capacity dashboards (e.g., [4], [5], [12])	Regional and multi-hospital networks	Reduces localized overload and improves system-wide resilience through coordinated decision-making	Dependent on governance alignment, interoperability, and timely data sharing; influenced by regulatory constraints	Implemented during large-scale outbreaks but varies across regions
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6. Implementation Challenges and Open Issues

Although forecasting, optimization, and coordination frameworks offer structured approaches to managing hospital load during epidemic outbreaks, their practical implementation remains complex. Abdel-Rahman et al. [22], Feda et al. [12], and Gupta et al. [3] highlight that operational, technical, ethical, and regulatory barriers continue to limit full-scale adoption in real-world healthcare environments. This indicates that developing analytical models is only one part of the challenge, while integrating them into practice introduces additional constraints.

A major issue relates to data quality and availability. Garber and Okhrin [2] emphasize that predictive and optimization models depend on accurate and timely datasets, while Mebrahtu et al. [27] show that hospital information systems often operate in fragmented silos with inconsistent data standards. Alsumait et al. [13] further note that missing values, delayed updates, and incompatible electronic health record systems can degrade model reliability. In practice, even small data inconsistencies can produce misleading forecasts or allocation errors, particularly during outbreak conditions where rapid decision-making is required.

Model robustness under dynamic outbreak conditions presents another challenge. Wohlfender et al. [16] show that many analytical frameworks are developed using retrospective datasets collected under relatively stable conditions, whereas Abdel-Rahman et al. [22] highlight that epidemic environments are highly dynamic, with evolving infection rates and policy interventions. This creates a gap between model

assumptions and real-world conditions. As a result, models that perform well on historical data may struggle to adapt to sudden shifts in healthcare demand, raising concerns about generalizability during real-time crisis management.

Interpretability and trust also play a critical role in implementation. Feda et al. [12] and Alsumait et al. [8] note that complex machine learning and optimization models may produce accurate outputs, but their adoption depends on whether clinicians and administrators understand and trust the results. Garber and Okhrin [2] further emphasize that decision-support systems must align with clinical judgment to be effective. This is important because models perceived as “black boxes” may face resistance, even when their performance is strong.

Privacy and data-sharing constraints further complicate regional coordination. Pett et al. [5] show that cross-hospital load balancing requires the exchange of operational and patient-level data, while Gupta et al. [3] highlight that regulatory frameworks and institutional boundaries often limit such data sharing. In many cases, governance fragmentation and legal restrictions delay or prevent the implementation of shared dashboards and coordinated transfer systems. This means that technical feasibility alone does not guarantee successful coordination.

Ethical considerations introduce an additional layer of complexity. Pett et al. [5] and Leeftink et al. [14] highlight that resource allocation decisions during severe surges may involve prioritizing patients for intensive care, transfers, or specialized treatment. While optimization models aim to improve efficiency, they must operate within ethical and regulatory boundaries. Questions related to fairness, equity, and accountability cannot be resolved through algorithmic design alone, which limits the direct application of purely optimization-driven approaches.

Scalability and sustainability beyond crisis periods remain open issues. Feda et al. [12] note that some hospitals adopt surge-management tools during emergencies but discontinue their use once demand stabilizes, while Alsumait et al. [8] highlight that this episodic implementation reduces long-term system learning. Mebrahtu et al. [27] suggest that sustainable impact requires embedding analytical tools into routine workflows rather than treating them as temporary solutions. This suggests that long-term integration is as important as short-term performance.

Overall, these challenges show that technological sophistication alone is insufficient to ensure effective epidemic load management. Abdel-Rahman et al. [22] and Feda et al. [12] demonstrate that successful implementation depends on data governance, model transparency, regulatory alignment, and institutional collaboration. Addressing these open issues is therefore essential for translating analytical research into resilient and operational healthcare systems.

7. Positioning of the Proposed Project

The reviewed literature demonstrates substantial progress in forecasting hospital demand, optimizing resource allocation, and enabling coordination across healthcare systems during epidemic surges. Studies such as Garber and Okhrin [2] and Wohlfender et al. [16] show that predictive models can estimate short-term admissions and occupancy with reasonable accuracy, while Abdel-Rahman et al. [7] and Parker et al. [22] propose optimization frameworks for allocating constrained resources. In parallel, Franklin et al. [27] and Feda et al. [12] highlight the role of command centers and shared dashboards in improving coordination. However, these approaches are typically developed and evaluated independently, which limits their effectiveness when applied to real operational decision-making.

A key limitation across the literature is the lack of integration between prediction, decision-making, and coordination. Forecasting models provide estimates of demand but do not specify how hospitals should respond when capacity thresholds are exceeded. Optimization models, on the other hand, often assume predefined demand inputs and therefore do not adapt dynamically to predicted changes. Coordination frameworks rely on institutional processes and shared visibility, yet they frequently lack analytical mechanisms that translate system data into actionable recommendations. As a result, existing approaches address individual components of surge management rather than supporting a continuous, end-to-end decision-support workflow.

To address this gap, the proposed project implements a prototype system that integrates predictive analytics, rule-based decision support, simulation, and cross-hospital coordination within a single framework. The system is developed as a simulated decision-support environment using hospital-level operational data.

These inputs include daily admissions, discharges, total bed occupancy, ICU usage, and predefined capacity limits, which together capture the dynamics of hospital load and patient flow.

The forecasting component is implemented using both statistical and machine learning approaches. A baseline time-series model (e.g., ARIMA) is used to capture temporal trends in admissions and occupancy, while a machine learning model such as Random Forest or XGBoost captures nonlinear patterns in the data. The system generates predictions for next-day demand as well as short-term trends over a 7-day horizon. These forecasts form the basis for detecting future overload risks and supporting proactive decision-making.

A rule-based decision-support module is incorporated to translate predictive outputs into operational actions. At the individual hospital level, overload conditions are identified by comparing predicted occupancy against predefined capacity thresholds, triggering recommendations such as increasing discharge prioritization, activating surge capacity, or adjusting resource allocation. At the network level, capacity conditions across hospitals are compared to identify imbalances, enabling recommendations for patient transfers or load redistribution. This ensures that decision-making addresses both internal hospital management and cross-hospital coordination. For example, if predicted occupancy exceeds capacity (e.g., 105 beds required for a 100-bed hospital), the system may recommend increasing discharge rates or activating additional capacity, while also suggesting patient transfers to nearby hospitals with available beds.

To evaluate these decisions, a simulation component is implemented at both the individual hospital level and across the hospital network. Internal simulations assess how actions such as increased discharge rates or temporary capacity expansion affect occupancy levels. In parallel, cross-hospital simulations model patient transfers between facilities and update capacity conditions accordingly. By comparing system states before and after applying recommendations, the simulation enables evaluation of “what-if” scenarios and supports assessment of decision effectiveness.

A centralized dashboard is used to present current and predicted hospital status, alerts, recommended actions, and simulation results. These outputs include both internal hospital scenarios (e.g., changes in discharge rates or temporary capacity expansion) and cross-hospital coordination outcomes (e.g., patient transfers between facilities). The interface provides both single-hospital views and multi-hospital network comparisons, allowing users to assess local capacity conditions alongside system-wide imbalances. It is

designed for coordination-level decision-makers and operates on aggregated operational data without including patient-level information, ensuring privacy and alignment with practical data constraints.

The system is evaluated using both predictive and operational performance metrics. Forecasting accuracy is assessed using standard measures such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). In addition, system-level performance is evaluated by analyzing the reduction of predicted overload events and improvements in capacity utilization after applying recommended actions within the simulation environment.

It is important to note that the proposed system is designed as a prototype for evaluation purposes rather than a deployable healthcare solution. The implementation relies on retrospective or publicly available datasets, supplemented by simulated data where necessary. Real-time integration with hospital information systems, regulatory compliance, and patient-level decision-making are outside the scope of this project. The focus is on demonstrating the feasibility and value of integrating forecasting, decision support, simulation, and coordination within a single analytical framework.

This integrated approach provides a practical and implementable framework for improving hospital load management under epidemic conditions.

8. Conclusion

The literature reviewed in this study demonstrates that managing hospital capacity during epidemic outbreaks requires more than reactive operational adjustments. Studies such as Garber and Okhrin [2], Abdel-Rahman et al. [7], and Parker et al. [22] show that predictive modeling, resource allocation, and coordination mechanisms each play a critical role in addressing surge conditions. Forecasting approaches improve the ability to anticipate demand, optimization techniques provide structured strategies for allocating limited resources, and coordination frameworks support load balancing across healthcare institutions. However, these contributions are typically developed within separate analytical streams.

A consistent finding across the reviewed studies is that this separation limits practical effectiveness. Forecasting models often stop at prediction without supporting operational decisions, while optimization and simulation approaches rely on predefined or static demand assumptions. Similarly, coordination frameworks emphasize governance and communication but do not always incorporate analytical mechanisms that guide real-time decision-making. This fragmentation makes it difficult for healthcare systems to translate analytical insights into coordinated operational responses during rapidly evolving outbreaks.

These observations suggest that the main challenge is not the absence of analytical tools, but the lack of integration between them. Connecting predictive outputs with actionable decision-support logic and coordination mechanisms is essential for improving system-level responsiveness. In practice, hospital operations require continuous interaction between demand estimation, resource allocation, and inter-hospital collaboration, rather than treating these components independently.

Based on these gaps, the proposed project focuses on developing a prototype decision-support framework that integrates predictive analytics, rule-based resource allocation, and simulated coordination within a unified system. By applying this framework to retrospective or simulated hospital data, the project demonstrates how predicted demand can be directly linked to operational recommendations and evaluated through simulation. This provides a concrete example of how fragmented analytical approaches can be combined into a coherent decision-support process.

Overall, this work shows that integrating forecasting, decision support, and coordination within a single framework offers a more practical approach to managing hospital load during epidemic conditions. Such an approach is not only aligned with the needs identified in the literature but also provides a feasible direction for developing operational tools that support healthcare system resilience under high-demand scenarios.

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